EA

ELITE IAL MATHEMATICS

EDEXCEL INTERNATIONAL A LEVEL

WME01/01 May/June 2023 Worked Solutions

May/June 2023 paper rebuilt with the worked solution after each question.

8
paper questions

WME01
Mechanics 1

Single-paper solutions
standalone IAL route

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Prepared for Dr Eslam Ahmed - eliteigcse.com

M1

PAST PAPER

WME01/01 May/June 2023

May/June 2023 | 8 questions | 75 marks

8
questions

75
marks

Answers
worked solution after each question

WME01/01 MAY/JUNE 2023

7 marks

Question 1

Momentum, Impulse & Collisions

1. A particle A has mass 4 kg and a particle B has mass 2 kg.

The particles move towards each other in opposite directions along the same straight line on a smooth horizontal table and collide directly.

Immediately before the collision, the speed of A is $2u\,\text{m s}^{-1}$ and the speed of B is $3u\,\text{m s}^{-1}$

Immediately after the collision, the speed of B is $2u\,\text{m s}^{-1}$

The direction of motion of B is reversed by the collision.

(a) Find, in terms of u, the speed of A immediately after the collision.

(3)

(b) State the direction of motion of A immediately after the collision.

(1)

(c) Find, in terms of u, the magnitude of the impulse received by B in the collision. State the units of your answer.

(3)

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Worked Solution - Question 1

Momentum, Impulse & Collisions

Topic group

1. Choose a positive direction

Take the original direction of A as positive. Before the collision, A has velocity $2u$ and B has velocity $-3u$. After the collision, B has velocity $2u$ because its direction is reversed.

2. Use conservation of momentum

Let the velocity of A after the collision be v. Then $4(2u) + 2(-3u) = 4v + 2(2u)$.

3. Solve for the velocity of A

$8u - 6u = 4v + 4u$, so $2u = 4v + 4u$. Hence $4v = -2u$ and $v = -\frac{u}{2}$.

4. State the speed and direction

The speed of A is the magnitude of its velocity, so it is $\frac{u}{2}$. The negative sign shows that A has reversed direction.

5. Find the impulse on B

Impulse equals change in momentum. For B,
$I = 2\left(2u - (-3u)\right) = 2(5u) = 10u$.

6. Attach the units

The impulse is $10u\ \mathrm{N\,s}$, equivalently $10u\ \mathrm{kg\,m\,s^{-1}}$.

Final answer

(a) speed of A $= \frac{u}{2}$. $\quad (b)$ A reverses direction. $\quad (c)$ $I = 10u\ \mathrm{N\,s}$.

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10 marks

Question 2

Working with Vectors

[In this question **i** *and* **j** *are horizontal perpendicular unit vectors.]*

2. A particle P rests in equilibrium on a smooth horizontal plane.

A system of **three** forces, $\mathbf{F}_1$ N, $\mathbf{F}_2$ N and $\mathbf{F}_3$ N where

$$\mathbf{F}_1 = (3c\mathbf{i} + 4c\mathbf{j})$$
$$\mathbf{F}_2 = (-14\mathbf{i} + 7\mathbf{j})$$

is applied to P.

Given that P remains in equilibrium,

(a) find $\mathbf{F}_3$ in terms of c, **i** and **j**.

(2)

The force $\mathbf{F}_3$ is **removed** from the system.

Given that $c = 2$

(b) find the size of the angle between the direction of **i** and the direction of the resultant force acting on P.

(4)

The mass of P is m kg.

Given that the magnitude of the acceleration of P is $8.5\,\mathrm{m\,s^{-2}}$

(c) find the value of m.

(4)

Worked Solution - Question 2

Working with Vectors

Topic group

1. Use equilibrium

For equilibrium, $\mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 = \mathbf{0}$. Hence $\mathbf{F}_3 = -(\mathbf{F}_1 + \mathbf{F}_2)$.

2. Find F3

$\mathbf{F}_1 + \mathbf{F}_2 = (3c - 14)\mathbf{i} + (4c + 7)\mathbf{j}$, so $\mathbf{F}_3 = (14 - 3c)\mathbf{i} + (-7 - 4c)\mathbf{j}$.

3. Find the resultant when c = 2

When $c = 2$, $\mathbf{F}_1 = 6\mathbf{i} + 8\mathbf{j}$ and $\mathbf{F}_2 = -14\mathbf{i} + 7\mathbf{j}$. Therefore the resultant is $-8\mathbf{i} + 15\mathbf{j}$.

4. Find the angle

The resultant is in the second quadrant. Its angle from the positive $\mathbf{i}$ direction is $180° - \tan^{-1}\left(\frac{15}{8}\right) = 118.1°$, so the angle is about $118°$.

5. Find the magnitude of the resultant

$|\mathbf{R}| = \sqrt{(-8)^2 + 15^2} = 17\ \mathrm{N}$.

6. Use F = ma

The acceleration magnitude is $8.5\ \mathrm{m\,s^{-2}}$, so $17 = 8.5m$. Hence $m = 2$.

Final answer

$(a)\ \mathbf{F}_3 = (14 - 3c)\mathbf{i} + (-7 - 4c)\mathbf{j}. \quad (b)\ 118°\text{ approximately.} \quad (c)\ m = 2$

.

8 marks

Question 3

Constant Acceleration in 1D

3. Two students observe a book of mass 0.2 kg fall vertically from rest from a shelf that is 1.5 m above the floor.

Student *A* suggests that the book is modelled as a particle falling freely under gravity.

(a) Use student *A*'s model to find the time taken for the book to reach the floor.

(3)

Student *B* suggests an improved model where the book is modelled as a particle experiencing a constant resistance to motion of magnitude *R* newtons.

Given that the time taken for the book to reach the floor is 0.6 seconds,

(b) use student *B*'s model to find the value of *R*

(5)

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Worked Solution - Question 3

Constant Acceleration in 1D

Topic group

1. Use free fall for student A

From rest, with downward displacement 1.5 m and acceleration g, use $s = ut + \frac{1}{2}at^2$.

2. Find the time

$1.5 = 0 + \frac{1}{2}(9.8)t^2$, so $t^2 = \frac{3}{9.8}$ and $t = 0.553\ldots$ s.

3. Find the acceleration in student B's model

Now the time is 0.6 s. Again starting from rest, $1.5 = \frac{1}{2}a(0.6)^2 = 0.18a$, so $a = 8.333\ldots\ \mathrm{m\,s^{-2}}$ downward.

4. Apply Newtons second law

Taking downward as positive, weight is $0.2g$ and resistance R acts upward. Therefore $0.2g - R = 0.2a$.

5. Find R

$1.96 - R = 0.2(8.333\ldots) = 1.666\ldots$, so $R = 0.293\ldots$ N.

Final answer

$(a)\ t = 0.553$ s approximately. $\quad (b)\ R = 0.293$ N.

Question 4

12 marks

Moments

4.

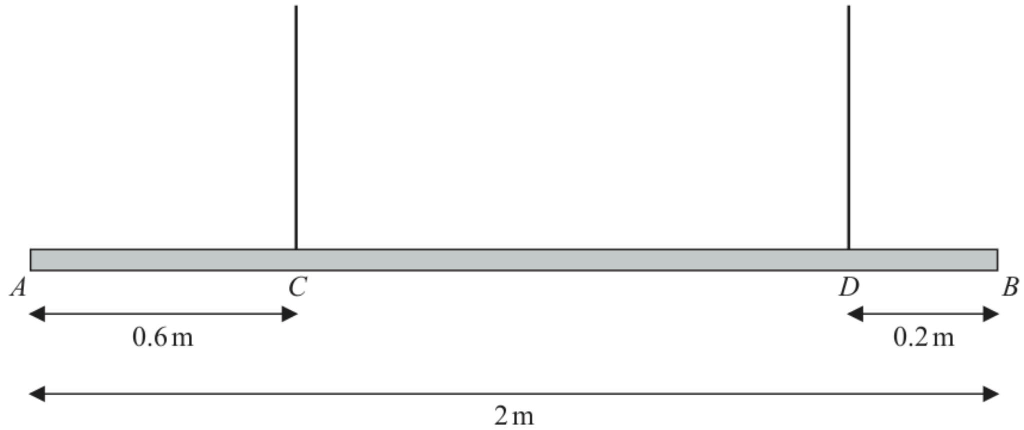

Figure 1

Figure 1 shows a beam AB, of mass m kg and length 2 m, suspended by two light vertical ropes.
The ropes are attached to the points C and D on the beam, where $AC = 0.6$ m and $DB = 0.2$ m
The beam is in equilibrium in a horizontal position.

A particle of mass pm kg is attached to the beam at A and the beam remains in equilibrium in a horizontal position.

The beam is modelled as a uniform rod.

(a) Given that the tension in the rope attached at C is four times the tension in the rope attached at D, use the model to find the exact value of p.

(7)

The particle of mass pm kg at A is removed and replaced by a particle of mass qm kg at A.
The beam remains in equilibrium in a horizontal position but is now on the point of tilting.

(b) Using the model, find the exact value of q

(4)

(c) State how you have used the modelling assumption that the beam is uniform.

(1)

Worked Solution - Question 4

Moments

Topic group

1. Set the geometry

The beam is 2 m long. Since $AC = 0.6$ m and $DB = 0.2$ m, point D is 1.8 m from A. The uniform beam has weight mg acting 1.0 m from A.

2. Name the tensions

Let the tension at D be T. The tension at C is four times this, so it is $4T$.

3. Use vertical equilibrium

$4T + T = pmg + mg$, so $5T = (p+1)mg$.

4. Take moments about A

The particle at A has no moment about A. Hence $4T(0.6) + T(1.8) = mg(1.0)$.

5. Find T

$2.4T + 1.8T = mg$, so $4.2T = mg$ and $T = \dfrac{5mg}{21}$.

6. Find p

Substitute into $5T = (p+1)mg$: $5 \cdot \dfrac{5mg}{21} = (p+1)mg$. Thus $p + 1 = \dfrac{25}{21}$ and $p = \dfrac{4}{21}$.

7. Use the tilting condition

When the particle of mass qm is at A and the beam is on the point of tilting, the rope at D is just slack, so its tension is zero.

8. Take moments about C

About C, the particle at A has moment $qmg(0.6)$ and the beam weight has moment $mg(0.4)$ in the opposite direction. Thus $qmg(0.6) = mg(0.4)$, giving $q = \dfrac{2}{3}$.

9. Use the modelling assumption

The beam is uniform, so its centre of mass is at the midpoint of the beam and its weight acts there.

Final answer

(a) $p = \dfrac{4}{21}$. (b) $q = \dfrac{2}{3}$. (c) The beam weight acts at its midpoint.

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Question 5

11 marks

Kinematics Graphs

5.

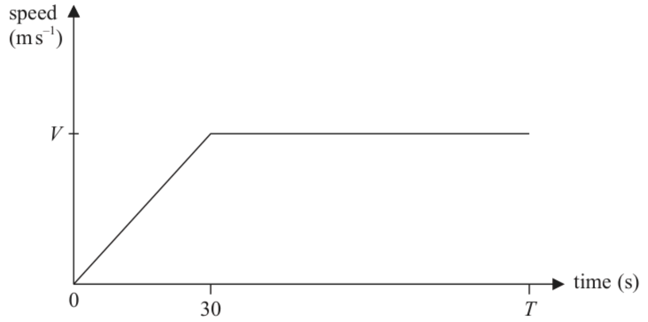

Figure 2

The speed-time graph in Figure 2 illustrates the motion of a car travelling along a straight horizontal road.
At time $t = 0$, the car starts from rest and accelerates uniformly for 30 s until it reaches a speed of $V\,\mathrm{m\,s^{-1}}$
The car then travels at a constant speed of $V\,\mathrm{m\,s^{-1}}$ until time $t = T$ seconds.

(a) Show that the distance travelled by the car between $t = 0$ and $t = T$ seconds is $V(T - 15)$ metres.

(2)

A motorbike also travels along the same road.

- The motorbike starts from rest at time $\boldsymbol{t} = \mathbf{10\,s}$ and accelerates uniformly for 40 s
- The acceleration of the motorbike is the **same** as the acceleration of the car
- The motorbike then travels at a constant speed for a further 10 s before decelerating uniformly until it reaches a speed of $V\,\mathrm{m\,s^{-1}}$ at time T seconds

(b) On Figure 2, sketch a speed-time graph for the motion of the motorbike.

[*If you need to redraw your sketch, there is a copy of Figure 2 on page 15.*]

(2)

(c) Show that the constant speed of the motorbike is $\frac{4V}{3}\,\mathrm{m\,s^{-1}}$

(2)

At time $t = T$ seconds, the distance travelled by each vehicle is the same.

(d) Find the value of T

(5)

Only use this copy of Figure 2 if you need to redraw your sketch.

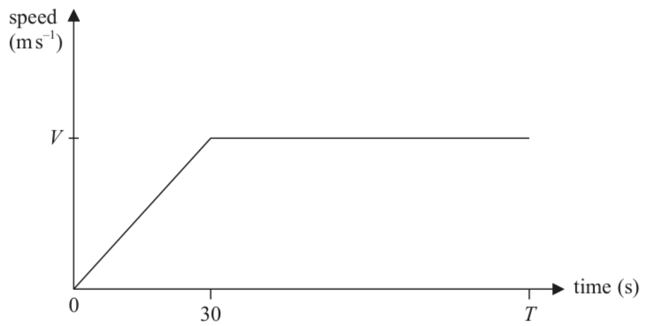

Figure 2

(Total for Question 5 is 11 marks)

Worked Solution - Question 5

1. Find the car distance

The car graph is a triangle from 0 to 30 plus a rectangle from 30 to T. Distance $= \frac{1}{2}(30)V + V(T-30)$.

2. Simplify the car distance

$\frac{1}{2}(30)V + V(T-30) = 15V + VT - 30V = V(T-15)$, as required.

3. Describe the motorbike graph

The motorbike starts at $(10, 0)$, accelerates with the same gradient as the car until $t = 50$, then travels at constant speed until $t = 60$, then decelerates linearly to (T, V).

4. Find the motorbike constant speed

The car acceleration is $\frac{V}{30}$. The motorbike accelerates for 40 s, so its speed is $40 \cdot \frac{V}{30} = \frac{4V}{3}$.

5. Find the motorbike distance

Its distance is $\frac{1}{2}(40)\left(\frac{4V}{3}\right) + 10\left(\frac{4V}{3}\right) + \frac{1}{2}\left(\frac{4V}{3} + V\right)(T-60)$.

6. Simplify the motorbike distance

This becomes $\frac{80V}{3} + \frac{40V}{3} + \frac{7V}{6}(T-60) = 40V + \frac{7V}{6}(T-60) = \frac{7VT}{6} - 30V$.

7. Equate the two distances

At time T, the distances are equal: $V(T-15) = \frac{7VT}{6} - 30V$. Divide by V to get $T - 15 = \frac{7T}{6} - 30$.

8. Find T

Multiplying by 6 gives $6T - 90 = 7T - 180$, so $T = 90$.

Final answer

(a) car distance $= V(T - 15)$.

(b) Motorbike graph starts at $t = 10$, reaches $\dfrac{4V}{3}$ at $t = 50$, stays constant to $t = 60$, then falls to V at $t = T$.

(c) constant speed $= \dfrac{4V}{3}$. $\quad (d)$ $T = 90$.

Question 6

7 marks

Forces

6.

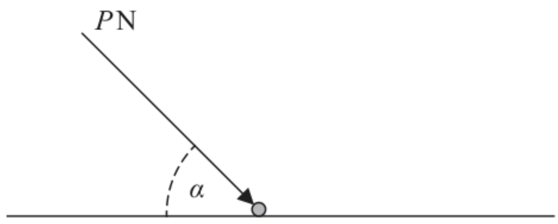

Figure 3

A particle of weight W newtons lies at rest on a rough horizontal surface, as shown in Figure 3.
A force of magnitude P newtons is applied to the particle.

The force acts at an angle α to the horizontal, where $\tan\alpha = \frac{4}{3}$

The coefficient of friction between the particle and the surface is $\frac{1}{4}$

Given that the particle does not move, show that

$$P \leqslant \frac{5W}{8}$$

(7)

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Worked Solution - Question 6

Forces

Topic group

1. Use the trig ratios

Given $\tan\alpha = \frac{4}{3}$, use $\sin\alpha = \frac{4}{5}$ and $\cos\alpha = \frac{3}{5}$.

2. Resolve vertically

The applied force increases the normal reaction in this model, so $R = W + P\sin\alpha$.

3. Resolve horizontally

For the particle not to move, friction must be able to balance the horizontal component of the applied force, so $P\cos\alpha \le F_{\max}$.

4. Use limiting friction

Since $\mu = \frac{1}{4}$, $F_{\max} = \frac{1}{4}R = \frac{1}{4}(W + P\sin\alpha)$.

5. Substitute the exact trig values

Thus $P \cdot \frac{3}{5} \le \frac{1}{4}\left(W + P \cdot \frac{4}{5}\right)$.

6. Rearrange the inequality

Multiplying by 20 gives $12P \le 5W + 4P$. Hence $8P \le 5W$, so $P \le \frac{5}{8}W$.

Final answer

$P \le \frac{5}{8}W$.

Question 7

11 marks

Newton's Second Law

7.

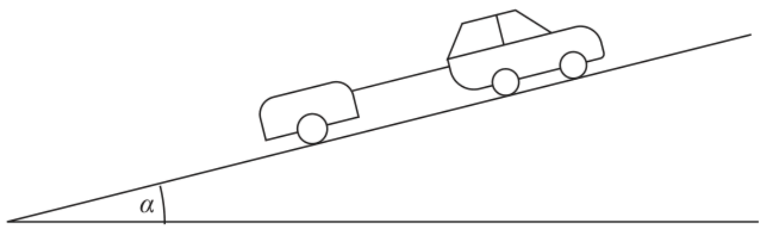

Figure 4

A car of mass 1200 kg is towing a trailer of mass 600 kg up a straight road, as shown in Figure 4.

The road is inclined at an angle α to the horizontal, where $\sin\alpha = \frac{1}{12}$

The driving force produced by the engine of the car is 3000 N.

The car moves with acceleration $0.75\,\text{m}\,\text{s}^{-2}$

The non-gravitational resistance to motion of

- the **car** is modelled as a constant force of magnitude $2R$ newtons
- the **trailer** is modelled as a constant force of magnitude R newtons

The car and the trailer are modelled as particles.
The tow bar between the car and trailer is modelled as a light rod that is parallel to the direction of motion.

Using the model,

(a) show that the value of R is 60

(4)

(b) find the tension in the tow bar.

(3)

When the car and trailer are moving at a speed of $12\,\text{m}\,\text{s}^{-1}$, the tow bar breaks.

Given that the non-gravitational resistance to motion of the trailer remains unchanged,

(c) use the model to find the further distance moved by the trailer before it first comes to rest.

(4)

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Worked Solution - Question 7

Newton's Second Law

Topic group

1. Use the whole system

For car plus trailer, the total mass is 1800 kg and the total resistance is $2R + R = 3R$. Taking up the slope as positive: $3000 - 1800g\sin\alpha - 3R = 1800(0.75)$.

2. Show R is 60

Since $\sin\alpha = \dfrac{1}{12}$, $1800g\sin\alpha = 1800 \cdot 9.8 \cdot \dfrac{1}{12} = 1470$. So $3000 - 1470 - 3R = 1350$, giving $180 = 3R$ and $R = 60\ \mathrm{N}$.

3. Apply Newtons second law to the trailer

For the trailer while it is being towed, $T - 600g\sin\alpha - R = 600(0.75)$.

4. Find the tow-bar tension

$T - 600 \cdot 9.8 \cdot \dfrac{1}{12} - 60 = 450$, so $T - 490 - 60 = 450$ and $T = 1000\ \mathrm{N}$.

5. Find the trailer acceleration after the break

After the tow bar breaks, the trailer is still moving up the slope, but the component of weight and resistance act down the slope. Taking up the slope as positive: $-600g\sin\alpha - 60 = 600a$.

6. Simplify the acceleration

$-490 - 60 = 600a$, so $a = -\dfrac{550}{600} = -\dfrac{11}{12}\ \mathrm{m\,s^{-2}}$.

7. Find the stopping distance

With initial speed $12\ \mathrm{m\,s^{-1}}$ and final speed 0, use $v^2 = u^2 + 2as$: $0 = 12^2 + 2\left(-\dfrac{11}{12}\right)d$. Hence $d = \dfrac{864}{11} = 78.5\ldots\ \mathrm{m}$.

Final answer

(a) $R = 60$ N. (b) $T = 1000$ N. (c) $d = 78.5$ m approximately.

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Question 8

9 marks

Working with Vectors

8.

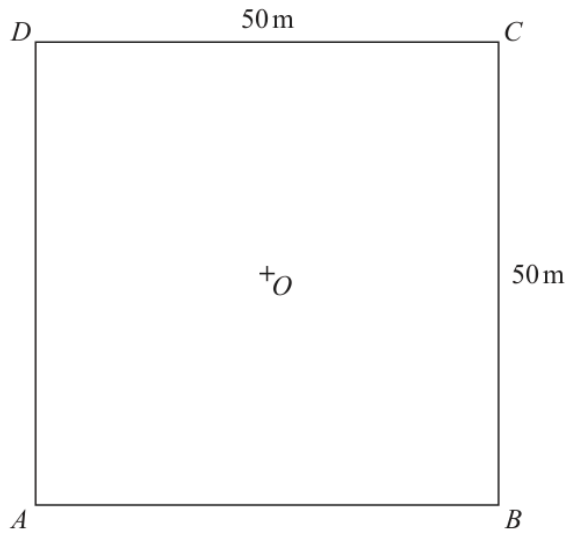

Figure 5

A square floor space $ABCD$, with centre O, is modelled as a flat horizontal surface measuring 50 m by 50 m, as shown in Figure 5.
The horizontal unit vectors $\mathbf{i}$ and $\mathbf{j}$ are in the direction of $\overrightarrow{AB}$ and $\overrightarrow{AD}$ respectively.

All position vectors are given relative to O.

A small robot R is programmed to travel across the floor at a constant velocity.

- At time $t = 0$, R is at the point with position vector $(-2\mathbf{i} + \mathbf{j})$ m
- At time $t = 11$ s, R is at the point with position vector $(9\mathbf{i} + 23\mathbf{j})$ m
- At time t seconds, the position vector of R is $\mathbf{r}$ metres

(a) Find, in terms of t, $\mathbf{i}$ and $\mathbf{j}$, an expression for $\mathbf{r}$

(3)

A second robot S is at the point C.

- At time $t = 0$, S leaves C and moves with constant velocity $(-\mathbf{i} - \mathbf{j})\,\text{m s}^{-1}$
- At time t seconds, the position vector of S is $\mathbf{s}$ metres

(b) Write down, in terms of t, $\mathbf{i}$ and $\mathbf{j}$, an expression for $\mathbf{s}$

(1)

(c) Show that

$$\overrightarrow{SR} = [(2t - 27)\mathbf{i} + (3t - 24)\mathbf{j}]\text{ m}$$

(2)

(d) Find the time when the distance between R and S is a minimum.

(3)

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Worked Solution - Question 8

Working with Vectors

Topic group

1. Find the velocity of R

R moves from $-2\mathbf{i}+\mathbf{j}$ to $9\mathbf{i}+23\mathbf{j}$ in 11 s. Its displacement is $11\mathbf{i}+22\mathbf{j}$, so its velocity is $\mathbf{i}+2\mathbf{j}$.

2. Write r

$\mathbf{r}=(-2\mathbf{i}+\mathbf{j})+t(\mathbf{i}+2\mathbf{j})=(t-2)\mathbf{i}+(2t+1)\mathbf{j}$.

3. Write s

Point C has position vector $25\mathbf{i}+25\mathbf{j}$. Since S moves with velocity $-\mathbf{i}-\mathbf{j}$, $\mathbf{s}=(25-t)\mathbf{i}+(25-t)\mathbf{j}$.

4. Find SR vector

$\overrightarrow{SR}=\mathbf{r}-\mathbf{s}=[(t-2)-(25-t)]\mathbf{i}+[(2t+1)-(25-t)]\mathbf{j}$.

5. Simplify SR

Therefore $\overrightarrow{SR}=(2t-27)\mathbf{i}+(3t-24)\mathbf{j}$ m.

6. Minimise the distance squared

The distance squared is $d^2=(2t-27)^2+(3t-24)^2$.

7. Expand the quadratic

$d^2=4t^2-108t+729+9t^2-144t+576=13t^2-252t+1305$.

8. Find the minimum time

The quadratic is minimum at $t=\dfrac{252}{2\cdot 13}=\dfrac{126}{13}=9.69\ldots$ s.

Final answer

(a) $\mathbf{r} = (t-2)\mathbf{i} + (2t+1)\mathbf{j}.$ (b) $\mathbf{s} = (25-t)\mathbf{i} + (25-t)\mathbf{j}.$ (c) $\overrightarrow{SR} = (2t-27)\mathbf{i} + (3t-24)\mathbf{j}.$ (d) $t = \dfrac{126}{13}\ \mathrm{s} \approx 9.7\ \mathrm{s}$

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